

Determination of Drought Triggering Probability via Bayesian Network

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Repo at <https://github.com/nishadhka/bn-ibf-e4drr>

Includes

- python code (well documented with plenty of docstrings)
- folder entitled `jupyter_notebooks` with 3 jupyter notebooks
- folder entitled `white-paper` with latex code and PDF for **preliminary** white-paper

<https://github.com/nishadhka/bn-ibf-e4drr/blob/main/white-paper/anticipatory-action.pdf>

I would appreciate it if you would take a look at the code and read the white paper. Comments from **why did you do this instead of that**, or **I think this part is wrong**, or **BS** or too technical or unnecessary or obscure are welcomed and I will attempt to address them.

Brief review of Tim Palmer ECMWF Theory

I^* = impact threshold for drought, it equals minus an SPI (Standardized Precipitation Index)

μ = ensemble member index

M = number of ensemble members, 51 for *ECMWF*

$\mathbb{1}()$ = indicator function. $\mathbb{1}(a > b)$ equals 1 if $a > b$ and is zero otherwise

TD = triggering decision to do drought warning

P^* = threshold probability that triggers a drought warning

$$\underbrace{P(\underline{I} \geq I^*)}_{\text{probability that impact is greater than threshold } I^*} = \frac{1}{M} \sum_{\mu=1}^M \mathbb{1}(I_{\mu} \geq I^*)$$

$$TD = \mathbb{1}[P(\underline{I} \geq I^*) > P^*]$$

Problem to solve

Find an automatic method of determining triggering probability P^* .

- The new method should be better than Nishadh's method of doing this for Karamoja, The latter is based solely on FAR/HR/AUROC.
- The new method should take into account **antecedents** like population density, road density, historical catastrophes, soil moisture, etc.

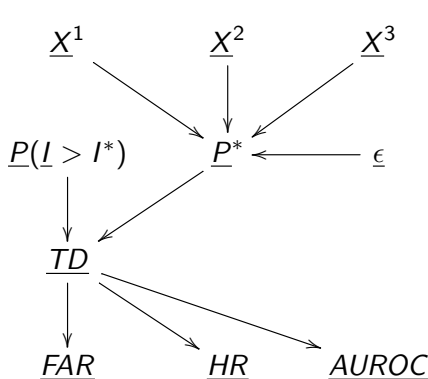
Bnet for determining triggering probability P^*

- **bnet (Bayesian network)**, discussed in my free, open source book

<https://github.com/rrtucci/Bayesuvius/raw/master/main.pdf>

- I do bnet calculations using **Pyagrum** software at

<https://pyagrum.readthedocs.io/en/> combined with a small amount of auxiliary code that I wrote.



$\underline{X}^1, \underline{X}^2, \underline{X}^3$ are antecedents
 $\underline{\epsilon}$ is noise

Use of Climada to find P^*

This project will use the Climada software library.

https://github.com/CLIMADA-project/climada_python

The table below relates their terminology to ours.

Climada terminology	Our terminology
Intensity of event	S =severity of event
Vulnerability= $\frac{Impact}{Exposure}$	$P_{danger} = \frac{IC}{RC}$ =danger probability
Impact	IC = insurance cost (\$)
Exposure	RC =replacement cost (\$)

In Climada, a hazard model yields the intensity for a particular event. Climada takes as input the intensity of an event, and calculates from this the event's vulnerability, impact and exposure. In this project, we use Climada for each flag $X^i \in \{0, 1\}$.

$$P^* = 1 - \max_i \{P_{danger}^i X_{\kappa}^i\}$$

Much Still to Do

Need Help

- **Need real data to train bnet.** So far, I've only used synthetic data to train it.
- Need to do a lot of work **fine tuning** and **integrating** this software with full corpus of DAA (Drought Anticipatory Action) software
- Need to do **modelling of antecedent probabilities**. This requires real data, which I haven't gotten yet.